Properties of inequality

1 18 is not greater than 23 or 18 is less than or equal to 23

2

a
- i $13 < 16$
- ii Yes

b
- i $16 > 13$
- ii Yes

c
- i $-8 < -5$
- ii Yes

d
- i $7 > 1$
- ii Yes

e
- i $-4 < -2$
- ii Yes

f
- i $9 > 7$
- ii Yes

g
- i $10 < 14$
- ii Yes

h
- i $-24 < -40$
- ii No

i
- i $\frac{5}{6} > \frac{2}{6}$
- ii Yes

j
- i $-\frac{8}{4} < -\frac{12}{4}$
- ii No

3 For $a < b$ to be true, a must lie to the left of b on a number line.
Since adding 1 to a number is equivalent to moving that number to the right on a number line, the distance between them on the line will remain the same, so $a + 1$ will still be to the left of $b + 1$.
This same logic applies for $a - 1 < b - 1$, and extends to the general case of $a + n < b + n$ as well.

4

a False　b False　c True　d True

5

a True　b True　c False　d True

6

a Yes　b No　c Yes　d Yes

7

a Yes　b Yes　c No　d Yes

8

a False　b True　c True　d False

$$
\begin{aligned}
a+a &< b+a \\
a+a+a &< b+a+a \\
3a &< b+a+a \\
3a &< b+b+b \\
3a &< 3b
\end{aligned}
$$